

D5 Standing Waves

The speed of sound in air is 330 m s^{-1} .

- D5.1 At an antinode of a standing wave on a string, at a particular time, the amplitude of the motion is 5.0 mm and the string is 3.5 mm above its equilibrium position.
For the string at the same time at an adjacent antinode,
- what is the amplitude?
 - what is the displacement?
- D5.2 Microwaves reflect from a metal surface. A detector moved away from the surface records alternating points of strong and weak signal. The distance between a place of strong signal and its nearest place of weak signal is 0.60 cm . Calculate the frequency of the microwaves.
- D5.3 The vibrating part of a violin string is 32.5 cm long. The ends of the string are fixed. The player touches the mid point of the string lightly, preventing the string moving side to side at that place.
- Is the touched point an antinode or a node?
 - What is the wavelength on the string of the lowest note which can be played this way?
 - If the speed of the wave on the string is 286 m s^{-1} calculate the wavelength of the sound in air.

CHAPTER D. WAVES

Answers

D5.1

- a) 5.0 mm amplitudes at all antinodes are the same
- b) -3.5 mm same displacement but in opposite direction

D5.2 Distance from node to nearest antinode = $\lambda/4 = 0.60$ cm.

Wavelength $\lambda = 2.40$ cm.

$$\text{Frequency } f = \frac{c}{\lambda} = \frac{3.0 \times 10^8}{0.024} = 12.5 \text{ GHz.}$$

D5.3

- a) node
- b) There are two node-node spacings along the string.
Each node-node spacing is $\lambda/2$.
Wavelength here will be the string length 32.5 cm.

$$\text{c) } f = \frac{c_{\text{string}}}{\lambda_{\text{string}}} = \frac{286}{0.325} = 880 \text{ Hz}$$
$$\lambda_{\text{air}} = \frac{c_{\text{air}}}{f} = \frac{330}{880} = 0.375 \text{ m}$$